Maksim Solovev

DevOps Engineer, 37 y.o.

13+ years in IT, 6+ years in DevOps · Russia, Moscow

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ABOUT ME

DevOps Expert Engineer. Running DevOps for the Jet Detective anti-fraud platform (Jet Infosystems). Built CI/CD, monitoring, and a management system for 160+ VMs from scratch.
Experience leading a team of 7, shipping releases to production on large government projects, and adapting services for Kubernetes.
I develop MCP tools for infrastructure and documentation used by both people and agents. I researched agent harnesses and local models for diagnostics in a closed network, testing quality on reproducible scenarios and identifying the limits of automation.

WORK EXPERIENCE

DevOps Expert Engineer

Oct 2023 – present

Jet Infosystems

Jet Detective — a fraud prevention and financial monitoring platform (anti-fraud for banks, retail, and industry). Responsible for all product infrastructure.

Infrastructure & automation

- Developed JD-Gateway (Python) — a single source of truth for 160+ VMs in vSphere: lifecycle management, TTL, REST API, dynamic inventory for Ansible and Prometheus, Web UI
- Created an automated product installer supporting CentOS, RedOS, and Astra Linux
- Automated VM provisioning via Packer (4 OSes) and Ansible (10+ roles)
- Built a backup system for PostgreSQL, Jenkins, OpenSearch, and Kafka on MinIO S3, with upload integrity checks, file archive checksum verification, and monitoring for stale backups and incomplete backup attempts
- Integrated Apache Superset into the product for embedded BI dashboards

CI/CD

- Designed and maintain complex Jenkins pipelines for build and deploy
- Implemented parallel builds and dynamic worker creation in vSphere
- Added auto-tests and SCA dependency scanning to the PR pipeline
- Automated product upgrade and rollback checks on temporary environments: VM creation, installation, version transitions, automated tests, and environment teardown
- Migrated repositories from Bitbucket to GitLab while preserving CI/CD integrations

Monitoring & logging

- Built the monitoring stack from scratch: Telegraf + Prometheus + Grafana + Thanos
- Set up alerting, created dashboards, implemented a flexible notification system
- Deployed OpenSearch + Loki + Promtail + Logstash for logs

Security

- Implemented HashiCorp Vault (GitOps) for secrets management
- Developed an SSH CA with Web UI and LDAP authentication
- Addressed security audit findings for a banking client

LLM & agents

- Researched an agent harness for routine DevOps work: compared frameworks, set up evaluations for grounding, security, and tool selection, and traced model calls; the experiment is now on hold
- Developed a PoC diagnostic agent using local models; tested reproducible scenarios to establish where models help with investigation and explanation, and where deterministic rules are needed
- MCP servers on top of documentation and infrastructure — shared context for the team and LLM agents

- MCP server on top of JD-Gateway — the agent spins environments up, shuts them down, and snapshots them itself, with no clicking around vSphere
- Vault MCP for agents: secrets served under deny policies, token isolated

Custom tooling

- Telegram bot for infrastructure management: LDAP, audit logging, brute-force protection
- JD release portal in Go: release cards with artifacts, Jira sync, LDAP authorization, and audit logging
- Documentation platform based on MkDocs with drift-detection

Team Lead DevOps

Dec 2019 – Oct 2023

Jet Infosystems

Projects: EGRN, Atlant (Testing Automation Platform). Led a DevOps team of 7.

EGRN project

- Test infrastructure: 6+ environments, 1000+ VMs
- Shipped releases to production — debugging under fire, log analysis
- Production MongoDB upgrade — playbooks, production-like environment, data migration

Atlant / TAP project

- System architect: design, implementation, and support of a test automation platform
- Adapted services for Kubernetes, deployed k8s clusters
- CI/CD consulting for internal projects, pre-sales for banking clients

System Administrator / Web Developer

Mar 2013 – Oct 2019

SaunaMaster → EOS Premium Spa Technologies

Started as the sole IT specialist in the company, grew from sysadmin to web developer.

Sysadmin (2013–2015)

- Designed small-business IT infrastructure from scratch: LAN, servers, IP telephony, CCTV

Web development (2015–2019)

- Commercial website development: DigitalOcean, Nginx, Python, Flask, MongoDB
- Built promotional sites for marketing campaigns

SELECTED CASES

01. JD-Gateway — single source of truth for 160+ VMs

Challenge: 160+ virtual machines across environments. Information scattered across Confluence, engineers' heads, and vSphere consoles.

Solution: Built JD-Gateway in Python: REST API and Web UI, vSphere integration, dynamic inventory for Ansible and Prometheus, TTL, registry checks to find untracked VMs, resource availability estimates for new environments, and tracking of IPs in use across environments.

Result: Creating a host used to take 30 minutes by hand through vSphere — now it's a couple of minutes. 14 people handle these operations themselves; the requests have stopped coming. TTL prevents environments from turning into zombies.

Lesson: *"Single source of truth" is not an architectural pattern — it's a matter of culture. The technical part is 30%; the rest is persuasion and convenience.*

02. Monitoring from scratch for an anti-fraud platform

Challenge: A product with zero monitoring. Issues were discovered when someone complained or an environment went down.

Solution: Built the full stack: Prometheus, Grafana, Thanos for long-term storage. Alerting with a flexible notification system. OpenSearch + Logstash for logs.

Result: Problems are visible before anyone complains. Dashboards became the primary diagnostic tool.

Lesson: *Monitoring isn't "install Prometheus". It's a dashboard that a specific person opens every morning. If it can't answer "is everything OK?" in 5 seconds — it's useless.*

03. Diagnostic agent on a local model

Challenge: Data from the bank's closed network can't go to the cloud. Simple question: can a local model handle the on-call role — go into the logs, metrics, and runbooks on its own, gather the facts, and reach a diagnosis.

Solution: Built a PoC with tools for logs, metrics, and runbooks, a registry of gathered facts, and answer validation. Compared local models with a cloud model on reproducible scenarios, then tested their behavior in a live environment.

Result: The local model helped gather and explain facts, but its assessment of system health was inconsistent: the same data produced different verdicts. The architectural conclusion was that rules should determine status and severity, while the model should assist with investigation.

Lesson: *A confident model response is no substitute for checking the facts.*

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04. Infrastructure you can just ask

Challenge: The team keeps coming to the DevOps engineer with the same questions: how does this pipeline work, how is the version computed, where do I look. The answers exist — but they're buried in Confluence, in Jenkinsfiles, and in my head. The classic "ask Maks".

Solution: Pulled the JD documentation into one place and put an MCP server with vector search on top of it. Both people and LLM agents query it: a specific question gets a specific answer with steps — no digging through ten pages.

Result: Live beta. Coverage is still growing, but some questions already go to the docs instead of to me. Agents pull context from there too — and make things up noticeably less.

Lesson: *It turned out not to be about the model. Boring docs aimed at a specific question beat a smarter model with no context — every single time.*

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05. Production MongoDB upgrade (EGRN)

Challenge: Upgrade MongoDB in production on one of the largest government IT projects. Downtime is critical, data loss is unacceptable.

Solution: Ansible playbooks for the upgrade. A production-like environment for full dry runs. Debugging until fully reproducible.

Result: Successful migration with zero data loss and minimal downtime.

Lesson: *In production there's no "let's try". Every step is verified on a staging environment. Playbooks are documentation that also executes.*

SKILLS

Core: Jenkins, GitLab CI/CD, Ansible, Docker, Prometheus, Grafana, HashiCorp Vault, VMware vSphere, Linux, PostgreSQL

Extended: Thanos, Packer, OpenSearch, Loki, Promtail, Logstash, Telegraf, Kafka, Keycloak, Nginx, Kubernetes, Terraform, Go, Python, Bash, Groovy, MinIO/S3, Apache Superset, Git

AI / LLM: MCP servers, Agent harnesses, LLM evals (promptfoo), LLM observability (Phoenix), Local inference, Agentic CLIs (Claude Code, Codex), OpenAI API

Previously: Flask, MongoDB, HTML/JS/CSS, DigitalOcean, SonarQube

LANGUAGES

Russian: native
English: advanced (reading, listening; working on speaking)

EDUCATION

Yandex Practicum — DevOps for Operations and Development (including Kubernetes) (2022–2023)

Pedagogical College No. 1 (K.D. Ushinsky) — Social Pedagogy (2006–2009)